

20 – Stage 2 Module Design Plan and Blueprints

How the nine application modules will be designed, in what order, to what template and against which review gates; with a blueprint for each module stating what it owns, what it exposes, the decisions already taken for it, and what it deliberately does not do.

Field	Value
Document	20 – Stage 2 Module Design Plan and Blueprints (design plan plus module-level blueprints; the detailed per-module designs are docs 21–29)
Version	v0.3 – draft for design review (adds M10 Procurement boundary; adds the implementation-guide layer 17A/19A/21A–29A and Sprint 0; re-plans the schedule; closes DR-1)
Date	8 September 2026
Status	Draft. Assumes doc 17 (Solution Architecture) module ownership as summarised in Section 2; doc 14 (Technical Foundations) and the sync contract are Stage 1 leftovers scheduled in the first two weeks of Stage 2.
Inputs	01–13 baseline v0.2 with the September change requests; 16 Questionnaire Answers; 18 Core Data Model; 19 Cross-Cutting Services; the September 2026 review decisions and the multi-MRP study; external reference: ARTS Operational Data Model and UnifiedPOS (NRF/OMG), GS1 general specifications, IAS 2, PDPA No. 9 of 2022, industry offline-first practice (Section 3).
Audience	Engineering leads, module designers, the federation technical reviewer. Sections 1–4 and 14 for programme readers.

Design stance: enterprise-grade, not enterprise-complicated. The system must carry 10,000 shops, 3 million receipts a day, multi-entity legal ownership, offline trading and an immutable audit trail. That demands rigour in a small number of places: ownership, numbering, immutability, sequencing and permissions. Everywhere else the plan chooses the plainest mechanism that meets the requirement, and Section 4 lists the mechanisms that were considered and deliberately left out. A module design that adds one of them back must justify it against that list.

1. What Stage 2 produces and how it runs

Aspect	Design
Output	For each module, a pair : the design document (docs 21–29, template in Section 5) and its implementation guide (21A–29A, outline in Section 5.1), handed to the module developer together. Sprint 0 delivers the two kernel guides (17A Build Skeleton, 19A Kernel Services) that every module guide inherits. Plus change requests against docs 18 and 19 where module work reveals a data-model or service gap.
Definition of done (per module)	Design: template complete; every phase-1 requirement has a design home; contract published; state machines reconciled with doc 03; data changes raised as change requests; test strategy sketched; prototype findings incorporated; decision requests listed with a decider. Guide: DDL, seed data and OpenAPI slice committed to the repository; handler specifications for every command; screen wireframes; test fixtures; ticket breakdown in build order; a developer can start the same day without asking a boundary question.
Cadence	Weekly design review (Thursday). A module enters review when its author declares it complete against the template; it exits when the reviewer signs the definition of done. Unresolved points become decision requests, never silent assumptions.
Team shape	Two to three module designers working in parallel tracks (Section 6), one architecture reviewer holding doc 17, and the till prototype engineer feeding findings in. Business reviewers are pulled in per module for state machines and screens-as-paper walkthroughs.
Duration	Re-planned in v0.3: Sprint 0 (17A, 19A) plus nine design-and-guide pairs over about nine weeks, 8 September to 6 November 2026 (Section 6). The blueprints' 70 design-days covered designs only; guides roughly double it. The pace is measured on 21A and re-planned once.
Change control	Module designs are drafts until the Stage 5 build-readiness review; but a published module contract (Section 5, interfaces) is frozen once a dependent module has started designing against it, and changes go through the same change request path.

2. Module map and boundary rules

Nine phase-1 application modules, plus the phase-2 Procurement boundary, sit on the ten kernel services of doc 19 v0.2. The list follows the Stage 1 plan with one adjustment: the trading documents that flow between entities (order, delivery note, GRN, discrepancy and claim, invoice, credit and debit notes, statements) form their own module rather than being split across inventory and billing. That is where the state machines the plan calls for actually live, and it keeps the ownership-transfer pivot (ADR-03) in one place.

#	Module	One-line purpose	Depends on
M1	Party, Tenancy & Security	Who exists, who owns what, who may do what	Cross-cutting only
M2	Catalogue & Batch	What is sold, in what units, under which barcodes; physical batches	M1
M3	Pricing & Rules	What it costs a buyer, what it sells for, every reduction, every ceiling	M1, M2
M4	Trading Documents	Order-to-settlement between entities; ownership transfer at GRN	M1, M2, M3
M5	Inventory, Costing, Repack & Loss	Where stock is, what it cost, how it moved, how it was lost	M1, M2, M4
M6	POS & Offline	The till: sell, receive, count, print, survive two days offline, sync	M1–M5
M7	Customers & Payment Recording	Registered customers, khata credit, tenders and payment receipts	M1, M4, M6
M8	Reporting Pipeline	Projections, aggregates, freshness stamps, exports, A4 rendering	All
M9	Integration	OpenAPI surface, journal mappings, e-invoice readiness, notifications	All
M10	Procurement (phase 2)	Supplier master, quotations, purchase orders, landed cost, supplier performance; boundary present from day one	M1, M2, M4, M5, M8

2.1 Boundary rules (the modular monolith contract)

- A module owns its tables (doc 18 assigns them) in its own schema namespace; no other module reads or writes those tables directly. Cross-module access is by in-process **query** interface (synchronous, read-only) or **command** interface (synchronous, transactional, permission-checked), or by reacting to **domain events** (asynchronous, via the outbox of doc 19).
- One database, one transaction: a command that spans modules runs in one transaction because it is one process against one database. No sagas, no compensation logic across modules in phase 1; compensation

exists only for *documents*, as the typed links of doc 18.

- Events are notifications of committed facts, not the source of truth. Module state lives in tables; the outbox lets other modules and tills react. No module is event-sourced.
- Each module publishes a contract file (commands, queries, events with versions). A dependent module designs against the contract, never against the tables. Contracts are the artefact that lets modules be designed in parallel.
- Module boundaries are also team boundaries for design ownership, but not deployment boundaries: one deployable, per doc 17, until measured load says otherwise.

3. Reference research applied

The module designs borrow from established references where those references settle a question cheaply, and deliberately not where they would import complexity the federation does not need.

Reference	What it is	How it is used here
ARTS Operational Data Model (NRF, now OMG)	The retail industry's reference relational model for transaction data, normalised for day-to-day retail operations, maintained since 1994 (source: OMG retail standards pages).	Vocabulary cross-check for M2–M7 (item, price derivation rule, retail transaction, tender, business unit). Not adopted as a schema: doc 18 is smaller and carries the federation's legal-entity split, which ARTS does not model.
UnifiedPOS (ARTS)	The platform-neutral specification for connecting POS peripherals (printers, scanners, scales, drawers, displays) to a terminal (source: ARTS/OMG).	M6's peripheral abstraction layer adopts UnifiedPOS device categories and state model as its interface vocabulary, without requiring UnifiedPOS-certified drivers on Android.
Offline-first practice	Current industry guidance (offline-first architecture guides and commercial offline POS products, 2025–2026) converges on: business record and outbox entry committed in one local transaction; an idempotency key on every mutation; a server-issued monotonic cursor for pulls rather than timestamps; and draft-then-permanent transaction models where the receipt number is assigned at completion. The acknowledged hard problem is reconciling two histories that diverged.	Confirms doc 19's outbox and sequencing design and doc 18's draft-then-issue protocol. The federation removes the hard problem entirely by the one-writer rule (ADR-10): histories never diverge because no record has two writers.
GS1 General Specifications and the 2D migration	Barcode syntaxes, application identifiers, in-store 20–29 prefixes, 2D readiness by 2027.	M2 barcode model and M6 scan layer (parse element strings and Digital Link; AI 01/10/17; dual marking).
IAS 2 Inventories	Weighted-average cost as an accepted cost formula; net realisable value.	M5 costing: per-entity moving weighted average across lots, NRV markdowns recorded as markdown rules (F-07), never as cost changes.
Consumer Affairs Authority Act No. 9 of 2003	Gazetted maximum prices operate from publication and bind all stock.	M3 control-price ceiling as a hard block; doc 18 control_price table.
Personal Data Protection Act No. 9 of 2022	Purpose limitation, minimisation, data subject rights.	M7 holds minimal customer data; phone as lookup with audited change and reuse detection (ADR-15, ADR-27).

4. Simplicity guardrails

Mechanisms considered and left out of phase 1. Each is a legitimate tool; each was judged to cost more than it returns for this federation. A module design that reintroduces one must argue against the stated reason.

Left out	Why	What is used instead
Microservices, separate deployables per module	Single DevOps operating model (ADR-20); one database is the strongest consistency tool available	Modular monolith with enforced boundaries (Section 2.1)
Event sourcing of aggregates	Reconstruction complexity, tooling, and no requirement it serves	Tables hold state; outbox events notify; audit log records
CRDTs / merge-based conflict resolution	Not needed: one writer per record type	ADR-10; sequence-ordered idempotent replay
Per-tenant schemas or databases	876 entities, cross-entity trading and a federation view make shared rows with RLS the natural fit	owner_entity_id + RLS policy classes (doc 18 3.7)
Workflow or BPM engine	Document lifecycles are fixed and few; an engine adds a product to operate	State machines in code, transitions in document_state_history
Rule-engine DSL / scripting for pricing	Cashier-safe rules need a closed vocabulary, not a language	Predicate and benefit JSON against a fixed schema (M3)
Distributed transactions, sagas	One database, one process	Single local transaction; document links for corrections
Kafka-class log broker in phase 1	Outbox already retains; broker only buffers and fans out	RabbitMQ or NATS (doc 19 DR-2)
Self-service report builder, GraphQL, API gateway product	Backlog by decision (O-05); REST with OpenAPI suffices for one expected consumer	Fixed catalogue reports; spreadsheet export; OpenAPI

Left out	Why	What is used instead
Plugin or extension framework for entities	Autonomy is served by configuration (roles, rules, price lists), not by code hooks	Configuration register (doc 07) as data
Shelf/back-store sub-locations, serialisation, returnable containers, cold-chain telemetry	Out of scope by answer (E-02, B-05, D-10, E-10)	–

5. Module design document template

Every module design (docs 21–29) follows this outline so that reviewers compare like with like and dependent designers find the contract in the same place.

Section	Content	Reviewer checks
1 Purpose and scope	One paragraph; phase-1 boundary; what is explicitly out	Matches the blueprint in this document
2 Requirements coverage	Table of every FR and NFR the module satisfies, with the design section that satisfies it	No phase-1 requirement of the module is missing
3 Domain model	Aggregates owned, invariants, references to doc 18 tables; change requests for anything new	No table owned by another module is written
4 State machines	For every document type the module owns: states, transitions, guards, permissions, audit events, offline-issuable transitions	Reconciled with doc 03; every transition has a permission and a reason rule
5 Contract	Commands (name, input, permission, transactional effect, events raised), queries (name, scope rules), events (type, version, payload)	Published in the contract file; versions assigned
6 Key flows	Sequence descriptions for the three to six flows that matter most, including the failure paths	Failure paths present; offline behaviour stated
7 Configuration	Every configurable item, its default, scope (federation/entity/location) and who may change it	Mapped to the configuration register
8 Screens as paper	Which paper document each screen mirrors; trilingual and low-literacy notes	Usability directive honoured
9 Non-functional design	Capacity, latency, offline, security, privacy specifics for this module	Traced to doc 06 NFR identifiers
10 Test strategy	Unit, contract, state-machine, property-based and scenario tests; the doc 13 scenarios the module must pass	Every invariant has a test named
11 Decision requests	Open points with decider and consequence	None silently assumed
12 Traceability	Requirement identifiers to sections	Complete

5.1 Implementation guide outline (docs nA)

The guide is self-contained: everything the developer needs from docs 14, 17–19, 30 and 32 is restated inline and marked *restated from doc X §Y*, so one document is read and provenance survives when a source changes. DDL, seed data and the OpenAPI slice are committed to the repository; the guide is their reviewed specification, never a second copy maintained by hand.

Section	Content
1 What you are building	One page: purpose, boundary, the module's place in the map; restated
2 What you inherit	The kernel classes, tables, shell components and conventions from 17A/19A the module uses, by name
3 Schema	Runnable DDL for the module's schema including constraints, RLS policies and indexes; seed data files
4 Package layout	Modulith module descriptor, package tree, naming, published package contents
5 API slice	OpenAPI for the module's commands and queries with request/response examples and error codes
6 Handler specifications	Per command: guard order, permission and MFA check, mutations, audit call, events raised, idempotency; pseudocode for the non-obvious ones
7 Queries and caches	Read paths, scope rules, per-instance caches and their invalidation
8 Screens	Wireframes per screen, shell components used, message ids, offline behaviour
9 Tests	Fixtures, the cases per layer, the property-based invariants, the RLS matrix rows for this module
10 Tickets	Build-order breakdown with estimates and the definition of done per ticket
11 Open points	Assumptions taken from pending decision requests, so the developer proceeds rather than waits

6. Sequence and parallel tracks

Dependency order fixes the start order, not a single file: once a module's contract is published, its dependants can begin. Three tracks run in parallel with the two Stage 1 leftovers front-loaded.

Week (2026)	Track A – back office core	Track B – edge and kernel	Track C – commercial and outward	Gates and inputs
1 8–11 Sep	21 M1 design (done); 21A begins once 19A conventions exist	Sprint 0: 17A Build Skeleton, 19A Kernel Services; doc 19 v0.2 and doc 18 v0.2 (done)	–	Hosting comparison due; tax advisor questions sent
2 14–18 Sep	21A M1 guide; M1 contract freeze	Sprint 0 continues; till spike 1 (local DB, counter, power loss)	–	Gate: 17A/19A reviewed; M1 contract frozen; pace of design+guide measured
3 21–25 Sep	22 M2 design and 22A	Sync contract conformance suite; spike 2 (Sinhala/Tamil thermal print); 26 M6 design begins against M1–M3 contracts	27 M7 design begins	Gate: M2 contract frozen; report catalogue workshop target
4 28 Sep–2 Oct	23 M3 design and 23A	26 continues; spike 3 (2D scan); cashier usability round 1	27A; 28 M8 design begins	Gate: M3 contract frozen; tax advisor answers expected
5 5–9 Oct	24 M4 design and 24A	26 review; 26A begins; spike 4 (sync load at envelope)	28A; 29 M9 design begins	Gate: M4 contract frozen; 24B numbering addendum if answered
6 12–16 Oct	25 M5 design and 25A	26A continues; usability round 2	29A	Gate: M5 contract frozen
7 19–23 Oct	Cross-module reconciliation; 33 API bundle assembled from the slices; CRs to 18/19	26A review	M7–M9 reviews	Gate: all contracts published
8–9 26 Oct–6 Nov (reserve)	Rework from advisor answers and spikes; 24B	Spike findings closed as ADRs	–	Build-readiness pre-check; Stage 4 may start

The schedule assumes three module designers plus the platform pair on Sprint 0 and the till track on M6. Each design-and-guide pair is estimated at roughly twice the design-only figure in its blueprint; the estimate is replaced by the measured pace of 21/21A at the week-2 gate.

Two rules keep the tracks honest: a track never designs against tables, only against a published contract; and a prototype finding that changes a contract is raised the same day, because a dependant may already be building on it.

7. M1 – Party, Tenancy & Security

Design document: 21 – Party, Tenancy and Security Design. **Estimated design effort:** 6 design-days.

Everything scopes on this module: who the legal parties are, which locations they own, which devices sit where, and who may do what. It is small, stable and must be right first, because every other module's row ownership and permission checks are its consumers.

7.1 Ownership

Owned aggregates and tables (doc 18)	Capabilities (SRS identifiers)
entity, entity_relationship, location, till_position, device, app_user, role, permission, role_permission, user_role; the RLS policy classes; the permission catalogue and role templates	FR-PTY-010–070 (parties, relationships, locations, positions); FR-SEC-010–040, 060, 070 (RBAC, authentication, sessions, segregation); FR-GOV-010–060 (visibility, self-administration, regulator access); NFR security set; ADR-18, ADR-26

7.2 Interfaces

Commands accepted	Queries served	Events published
RegisterEntity, OpenTradingRelationship, AmendRelationshipTerms, RegisterLocation, SetPrimaryTill, RegisterTillPosition, RetireTillPosition, EnrolDevice, AssignDeviceToPosition, SuspendDevice, CreateUser, DeactivateUser, ResetCredential, CreateRole, AmendRole, AssignRole, GrantExternalView	ResolveScope(user) → entity/location scopes and policy class; ResolvePermissions(user, rv); LookupRelationship(seller, buyer, date); ListLocations(entity); ListOperators(location) for till snapshot	entity.registered/updated.v1, relationship.opened/amended.v1, location.registered/updated.v1, till_position.registered/retired.v1, device.enrolled/assigned/suspended.v1, user.created/deactivated.v1, role.changed.v1, external_grant.issued/expired.v1

Depends on: Cross-cutting services only (doc 19).

7.3 Key design decisions

Decision	Rationale
Entity relationship carries the commercial terms, not the entity	Price list, credit limit, terms and dispute window are properties of a trading pair (A-06, A-07, A-09, D-06); putting them on the entity would force one set of terms per seller.
Shop is a location with a primary till	Locations hold no legal attributes; the primary till holds the location-scoped counters (GRN decision). Reassigning the primary till is a command with an audit event, not an edit.
Role templates clone on adoption; drift is notified, not forced	Entity autonomy (M-08); doc 19 DR-4 confirms.
Device suspension is central-immediate, device-eventual	An offline device cannot learn it is suspended; a signed revoke on next contact wipes caches. Stated and accepted rather than hidden.
Federation view is a policy class, not a super-user	Read everything, write nothing (M-02 to M-04, T-02). No Federation role maps to any entity-scoped write permission.

7.4 Deliberately not built

- No organisational hierarchy beyond entity → location; no departments, cost centres or regions as entities (district is an attribute).
- No self-service password reset in phase 1 (entity administrator resets; SMS reset is phase 2).
- No single sign-on with external providers.

7.5 Risks, prototypes and review focus

- Review focus: the RLS policy classes must be exercised by a test matrix of principal × table × operation before any other module depends on them.
- Risk: the 876-entity onboarding sequence is an operational process, not a design problem; the module must expose bulk registration for the onboarding playbook (S-07) without a special code path.

8. M2 – Catalogue & Batch

Design document: 22 – Catalogue and Batch Design. **Estimated design effort:** 6 design-days.

The shared product truth: SKU definition in one global namespace with local and shared status, units and pack conversions, barcodes many-to-one with optional batch resolution, tags as the grouping vocabulary, tax categories, and physical batches carrying expiry and printed MRP. It is the module the data steward operates.

8.1 Ownership

Owned aggregates and tables (doc 18)	Capabilities (SRS identifiers)
uom, sku, sku_uom_conversion, sku_barcode, sku_alias, sku_image, tag, sku_tag, tax_category, tax_rate, batch	FR-SKU-010–110 (definition, status, local creation, promotion with alias, barcodes, UOM, trilingual names, tags, tax category); B-01 to B-17 answers; ADR-07, ADR-08; FR-LOC-020 (trilingual master data); T-03 mitigations

8.2 Interfaces

Commands accepted	Queries served	Events published
CreateSku (draft), ActivateLocalSku, SubmitForSharing, PromoteToShared (with alias), DeactivateSku, DefineConversion, RegisterBarcode, RetireBarcode, RegisterBatch (from GRN or repack), CorrectBatchMrp (creates new batch, repoints lot, audited), DefineTag, TagSku, DefineTaxCategory, PublishTaxRate, AttachImage(sku barcode), RetireImage	LookupByBarcode(barcode, symbology, location) → sku, uom, optional batch; SearchSku(text, language, scope); GetCatalogueSnapshot(location, since_version) for tills (includes thumbnail keys for the location's assortment); GetThumbnail(key); ResolveAlias(sku); ListBatches(sku, location) with expiry	sku.created/activated/shared/promoted/deactivated.v1, barcode.registered/retired.v1, conversion.defined.v1, batch.registered/corrected.v1, tag.defined.v1, tax_rate.published.v1

Depends on: M1 (owner entity, scope). Batch registration is commanded by M4 (GRN) and M5 (repack).

8.3 Key design decisions

Decision	Rationale
SKU codes are opaque and system-generated; the English short name is the mandatory anchor	B-01, P-07. Meaningful codes rot; names translate.
Local SKUs live in the same namespace with status LOCAL and an owner	No separate ranges (B-09); visibility by owner and RLS; promotion keeps history via alias (B-10).
Barcode → batch is optional and owner-scoped	The hook that makes barcode-resolved MRP work (multi-MRP study, option B and C) without touching factory barcodes.
Batch is global; cost is on the lot	One physical batch, many successive owners at different costs (doc 18 DR-4).
Non-batch-tracked SKUs get a synthetic batch per receipt line	MRP and cost always have a home; price resolution has one code path (doc 18 DR-5).
Tags, not a hierarchy	B-15. Cross-entity comparison uses governed tags; entities may add local tags for their own merchandising.
Product images: one per SKU, optionally one per barcode, stored in object storage; tills cache thumbnails only	September 2026 decision. Low-literacy cashiers (P-08) and unbarcoded lookups benefit most from a picture. A per-barcode image covers pack-artwork changes, which usually coincide with a barcode change. Thumbnails (about 128 px, under 10 KB) ride in the catalogue snapshot for the location's assortment only, so sync cost stays bounded; a missing image falls back to a tag icon, never blocks a sale.

8.4 Deliberately not built

- No product attributes beyond name, description, units, tax and flags in phase 1 (no colour/size matrices, no variant families); a bounded entity-owned attributes jsonb on sku is the escape hatch (Appendix A).
- No supplier catalogue linkage in phase 1 (supplier master arrives with M9/procurement in phase 2; batch.supplier_id references a minimal supplier table owned here until then).
- No full-size images on the till, no images on receipts or customer displays, no image-based recognition.

8.5 Risks, prototypes and review focus

- Risk (programme, not design): the trusted catalogue assembly is the critical path (T-03). The module must ship the steward's tooling – bulk import with validation report, duplicate detection across three scripts, translation completeness dashboard – as first-class screens, not admin afterthoughts.
- Prototype input: the trigram-plus-ICU search behaviour on real Sinhala and Tamil product names (doc 19 section 5.4 test set) must be validated before M6 relies on it.

9. M3 – Pricing & Rules

Design document: 23 – Pricing and Rules Design. **Estimated design effort:** 8 design-days.

Every price a buyer pays and every reduction a customer receives, expressed as data the till can evaluate offline. Trade price lists per relationship, one retail list per MPCS, Federation advisory prices, the rule engine that is the sole source of any discount or markdown, and the two legal ceilings: printed MRP per batch and the gazetted control price by date.

9.1 Ownership

Owned aggregates and tables (doc 18)	Capabilities (SRS identifiers)
price_list, price_list_line, discount_rule, control_price; the price resolution algorithm (doc 18 section 4.9)	FR-PRC-010–130 (trade and retail lists, volume tiers, MRP enforcement, rules, markdowns, tax-inclusive pricing, rounding, per-customer preferential price P3); G-01 to G-10; F-07; ADR-14; the September 2026 multi-MRP decision and control-price finding

9.2 Interfaces

Commands accepted	Queries served	Events published
PublishTradePriceList(relationship, lines, effective_from), PublishRetailPriceList(entity, lines), PublishAdvisoryPrices(federation), AuthorRule, ActivateRule, RetireRule, EnterControlPrice, RescindControlPrice, SetMultiMrpPolicy(sku tag)	ResolvePrice(location, sku, uom, qty, batch?, date) → unit price, mrp_applied, control_price_applied, cap_reason; ApplyRules(basket) → benefits with rule ids; GetPricingSnapshot(location, since_version); CheckCeilingsForAuthoring(entity, sku, price)	price_list.published.v1, discount_rule.activated/retired.v1, control_price.entered/rescinded.v1 (all with apply_from), mrp_policy.changed.v1

Depends on: M1 (relationships, scope), M2 (SKU, batch MRP, tags, tax).

9.3 Key design decisions

Decision	Rationale
Closed rule vocabulary: predicate and benefit as JSON against a fixed schema	Rule kinds: time-limited price, quantity break, bill-level threshold, free item, days-to-expiry markdown. Predicates over sku/tag/qty/bill total/date/days-to-expiry only. No scripting; the till evaluates the same schema offline. Guardrail from Section 4.
Three-point MRP enforcement	At authoring (reject above any in-stock batch MRP or control price), at receipt (M4 raises a review when incoming batch MRP is below the standing list), at sale (M6 caps per line and records why). The September review split of FR-PRC-040.
Default multi-MRP policy AUTO_LOWEST; BARCODE_RESOLVED when the scan carries a batch; PICKER only by opt-in	The multi-MRP study recommendation: legally safe, fastest, invisible to the cashier; margin give-up bounded and measured via cap_reason.
Control price is a hard block at every enforcement point including label printing	CAA Act s.20(5): the gazette binds all stock from publication. Warnings would put the society at legal risk.
Every reduction stamps its rule id	ADR-14 and F-07: markdown cost is measurable separately from promotion cost; no discretionary discount exists anywhere.
Tax-inclusive retail prices computed from the rate in force; cash rounding at receipt level	G-09, G-10; line values stay exact.

9.4 Deliberately not built

- No scheduled future price lists in phase 1 (G-08 dropped); lists apply on propagation with apply_from at business-date granularity.
- No cross-entity promotion funding or settlement (G-06).
- No member or loyalty pricing of any kind (K-01); per-customer preferential price is phase 3 (FR-PRC-110).
- No price optimisation or elasticity modelling; price forecasting in procurement (phase 2) is a scenario tool, never an automated decision (L-02).

9.5 Risks, prototypes and review focus

- Review focus: the rule schema must be small enough to evaluate in a few milliseconds per basket on an ARM terminal and expressive enough for the doc 13 scenarios; a property-based test that no rule outcome can exceed a ceiling is mandatory.
- Risk: gazette currency. The module must expose the control-price register with an operational owner (doc 18 DR-8) and a staleness warning on tills.

10. M4 – Trading Documents

Design document: 24 – Trading Documents Design (order-to-settlement). **Estimated design effort:** 10 design-days.

The inter-entity flow: a buyer orders, the seller accepts and allocates, dispatches on a delivery note that may serve several drop points, the receiver counts and confirms a GRN at which moment ownership and risk transfer, discrepancies and claims lead to credit notes, the seller invoices and issues statements, and payments are recorded against specific invoices. This module owns the state machines the Stage 1 plan calls for; numbering detail waits for the tax advisor and is added as an addendum.

10.1 Ownership

Owned aggregates and tables (doc 18)	Capabilities (SRS identifiers)
document types ORD, DN, GRN, INV, CN, DN2, PRC (statement is a report); extension tables doc_order, doc_delivery_drop, doc_grn, doc_discrepancy, doc_claim; document_link usage for DISPUTES, CREDITS, DEBITS, SETTLES	FR-ORD-010–080 (requisition, suggestion input, approval, allocation FCFS with override, partial fulfilment and backorder, lock window, indicative ETA); FR-DEL-010–060 (delivery note, multi-drop, proof of delivery, actuals never forced, auto-discrepancy, photos, offline GRN); FR-BIL-010–150 (invoice, credit/debit notes, statements, matching to specific invoices, exposure warn-only, interest rule, e-invoice readiness); A-03 to A-13; C-01 to C-13; D-01 to D-09; I-05 to I-07; ADR-01, ADR-03, ADR-12

10.2 Interfaces

Commands accepted	Queries served	Events published
SubmitOrder, AcceptOrder(allocations, eta), AmendOrder (within lock), CancelOrder, IssueDeliveryNote(drops), RecordProofOfDelivery, CaptureGrn(actuals, photos) [offline], ConfirmGrn [offline], RaiseDiscrepancy (automatic), ResolveDiscrepancy, RaiseClaim, InspectClaim, ApproveClaim, IssueInvoice, IssueCreditNote, IssueDebitNote, RecordPaymentReceipt(settlements), OverrideLockWindow, EscalateToFederation	GetOrderStatus, ListOpenBackorders(seller), GetExposure(relationship) with freshness, ListUnsettledInvoices(relationship) for matching, GetStatement(relationship, period), GetDeliveryRun(vehicle, date)	order.submitted/accepted/amended/cancelled/locked.v1, delivery_note.issued/delivered.v1, grn.captured/confirmed.v1, discrepancy.raised/resolved.v1, claim.raised/approved/rejected.v1, invoice.issued.v1, credit_note.issued.v1, payment_receipt.recorded.v1, exposure.updated.v1

Depends on: M1 (relationships, terms), M2 (SKU, batch registration on GRN), M3 (trade price resolution at order and invoice). Publishes to M5 (stock effects of GRN and returns) and M7 (payment receipts, exposure).

10.3 Key design decisions

Decision	Rationale
GRN confirmation is the ownership pivot and is offline-issuable from a LOCATION series	ADR-03; H-02; the September GRN numbering decision. A receiver records actuals, never the dispatch note (D-04, D-05).
Discrepancy is raised automatically on any mismatch; credit is never automatic	D-05, D-07: the system records, the parties resolve, the Federation arbitrates on timeout (D-06).
Invoice follows GRN, not dispatch	Firm sale at GRN means the invoice is for what was received. Where the seller invoices on dispatch by agreement, the invoice references the DN and the GRN discrepancy drives a credit note; the design supports both, the relationship terms choose.
Payment receipts settle specific invoices via SETTLES links with amounts	I-07; exposure and ageing are computed from unsettled invoice amounts, and the system never processes money (ADR-01).
Exposure is informative: warn, never block	ADR-12. The warning is computed from manually recorded receipts and carries a freshness stamp.
Delivery note carries drops; vehicle, driver and timestamps are attributes, not a fleet module	D-01, D-02, D-09.

10.4 Deliberately not built

- No vehicle routing, load planning or fleet management (D-01).
- No automated interest posting (A-12: configurable default rule, manual application until accounting integration).

- No returnable container balances (D-10).
- No cross-entity promotional settlement; no pre-reserved number ranges (rejected, ADR-11).

10.5 Risks, prototypes and review focus

- Review focus: the six state machines (order, delivery note, GRN, discrepancy, claim, invoice/credit note) reconciled line by line with doc 03, with every transition's permission, reason rule and offline-issuability stated; the doc 13 scenarios 1 and 2 must pass on paper.
- Gate: numbering addendum after the tax advisor; DR: whether invoice-on-dispatch is offered as a relationship term in phase 1 or deferred.
- Risk: the Federation-as-arbiter flow is a governance process; the module should give it a small, explicit escalation state, not a workflow engine.

11. M5 – Inventory, Costing, Repack & Loss

Design document: 25 – Inventory, Costing, Repack and Loss Design. **Estimated design effort:** 8 design-days.

Where stock is and what it cost, by location and batch; the append-only movement ledger; weighted-average costing per entity; intra-entity transfers; repack recipes with expected yield; counts and adjustments with value-threshold approvals; and the write-off lifecycle with its loss categories, witness and photographic controls.

11.1 Ownership

Owned aggregates and tables (doc 18)	Capabilities (SRS identifiers)
stock_lot, stock_movement, repack_recipe; document types CNT, ADJ, WOF, RPK, XFR, OPB with extension tables; expiry warning tasks; reorder suggestions (P2)	FR-INV-010–100 (balances, movements, weighted average, FEFO, counts, cycle counting, adjustments with thresholds, negative stock with acknowledgement, ageing P2, reorder P2); FR-LOSS-010–060 (categories, ownership at loss, claims, approval by value, witness and photo, expiry alerts, markdown separation); E-01 to E-11; F-01 to F-11; B-06; L-06 (P3); R-02 (opening balances)

11.2 Interfaces

Commands accepted	Queries served	Events published
ApplyReceipt (from GRN), ApplySale/ApplySaleReversal (from receipts), TransferWithinEntity [XFR], ExecuteRepack(recipe, actual yield), ReverseRepack (while eligible), ScheduleCount, RecordCount [offline], ApproveAdjustment(thresholds), RequestWriteOff [offline draft], WitnessWriteOff, ApproveWriteOff, RecordOpeningBalance (two-party signed), AcknowledgeNegativeStock	GetBalance(location, sku) by batch with expiry; GetEntityAverageCost(entity, sku); ListExpiringBatches(location, window); GetAgeing(location) P2; GetLotSnapshot(location, since_version) for tills; GetLossSummary(scope, period, category)	stock.moved.v1, lot.negative.v1, count.recorded.v1, adjustment.approved.v1, writeoff.requested/witnessed/approved/posted.v1, repack.executed/reversed.v1, batch.expiring.v1, opening_balance.recorded.v1

Depends on: M1 (locations, thresholds by role parameters), M2 (SKU, batch, tags for thresholds), M4 (GRN and return stock effects). Consumed by M3 (in-stock batch MRPs for ceilings) and M6 (lot snapshots).

11.3 Key design decisions

Decision	Rationale
Movements are the only way stock changes; every movement cites a document	Doc 18 P4, B-I5. Balances are a materialised sum with a nightly reconciliation that raises, never silently fixes.
Weighted average per entity across lots, computed at issue	E-03, IAS 2. Lots keep acquisition cost for traceability; the entity average is the costing figure. No per-transaction override.
FEFO is the picking suggestion for expiry-tracked goods with manual override; it is not the pricing rule	E-04; the multi-MRP study separated rotation from pricing.
Repack consumes one SKU and produces another through a recipe with expected yield	ADR-07, B-06; variance measured against expectation; wrong-recipe correction is a reversal plus re-execution while the output lot is untouched (doc 13 scenario 3).
Write-off is the highest-fraud-risk lifecycle: requester ≠ approver, witness ≠ requester, photos, value-based approval	F-04, F-05, N-07; single-staff shops use the remote-witness rule (doc 18 DR-7).
Negative stock permitted with explicit acknowledgement and an exception event	E-09: blocking stops genuine sales whenever records are wrong.

11.4 Deliberately not built

- No shelf versus back-store locations (E-02); no bin or slot management in warehouses in phase 1.
- No system-calculated reorder points or ageing reports until phase 2 (E-08, E-11); the movement ledger is designed so both are pure projections later.
- No cold-chain telemetry (E-10); no disposal certificates (F-08).

11.5 Risks, prototypes and review focus

- Review focus: the costing arithmetic must be worked through the doc 13 scenarios with exact rupee results; a property-based test that lot quantities equal movement sums under random interleavings including reversals.
- Risk: opening balances at onboarding are the data-quality inflection point (R-02); the two-party signed opening balance document must be designed as a first-class flow, not a migration script.

12. M6 – POS & Offline

Design document: 26 – POS and Offline Design (till application and sync). **Estimated design effort:** 15 design-days.

The largest module and the one users touch: sell, refund within session, park and resume, weigh and price, cap to ceilings, apply rules, tender (recording only), print trilingual receipts as raster, deduct stock, receive goods, count, draft write-offs, switch to shop-management mode, and keep doing all of it for two days without central. Then sync: upload in sequence, download snapshots, never merge.

12.1 Ownership

Owned aggregates and tables (doc 18)	Capabilities (SRS identifiers)
document type RCT with extension doc_receipt_tender; till local database (same logical model as doc 18 for what it writes; read-only caches for what it receives); sync outbox and inbox; peripheral abstraction; receipt renderer; message catalogue on device	FR-POS-010–110 (sale, receipt, void, refund rules, park/resume, weighed items, negative-stock acknowledgement, offline set, cashier attribution, PIN, idle lock, shop-management mode); FR-PAY-010, 030–050 (tender recording, cash management: float, drops, blind close, variance); H-01 to H-13; I-01, I-03; N-03, N-09; O-02 (Z-report, day sales, till session offline); P-05; Q-07; ADR-10, ADR-19, ADR-24; the multi-MRP resolution policy

12.2 Interfaces

Commands accepted	Queries served	Events published
OpenTillSession(float), StartSale, AddLine(scan code weight), ResolveMrpPick (policy PICKER), AcknowledgeNegativeStock, ApplyRules, Tender(type, reference), CompleteSale (issue RCT), VoidSale, RefundSameSession, ParkSale/ResumeSale, CashDrop, CloseSession (blind), PrintZReport, SwitchToShopMode/ReturnToSales, and shop-mode delegates: CaptureGrn, ConfirmGrn, RecordCount, RequestWriteOff, ReprintDocument	Local: LookupBarcode, SearchSku, ResolvePrice, GetLot, GetPermissionSnapshot, ListParkedSales, GetSessionTotals. Central-facing: none; the till is a producer of events and a consumer of snapshots.	receipt.issued/voided/refunded.v1, till_session.opened/closed.v1, cash.dropped.v1, stock.moved.v1 (sale lines), mode.switched.v1, sync.batch_sent.v1, plus the M4/M5 events for shop-mode documents issued at the till

Depends on: Contracts of M1 (permission snapshot, operators), M2 (catalogue snapshot), M3 (pricing snapshot, ResolvePrice logic ported to device), M4 (GRN commands), M5 (lot snapshot, count and write-off commands); doc 19 event envelope; the sync contract.

12.3 Key design decisions

Decision	Rationale
Full local database, same logical model, one writer	H-01 (1–2 days offline); ADR-10. The till writes documents, lines, state rows, movements and audit; central pushes read-only caches. No merge code exists.
Draft-then-issue with the counter in the same local transaction	Doc 18 section 5.5; industry practice (Section 3). Power loss between steps leaves both or neither.
Snapshots apply atomically per version with apply_from at business-date granularity	A till never trades on a half-applied catalogue; all tills switch prices on the same day.
Peripheral abstraction modelled on UnifiedPOS categories	Scanner, POS printer, drawer, scale, line display, label printer (H-07) behind one interface; card terminals are standalone and only a reference is recorded (ADR-19).
Receipts render as raster through the platform text stack	P-05: correct Sinhala and Tamil on low-cost printers; English fallback per field.
Scan layer parses GS1 element strings and Digital Link; 2D imagers specified	Multi-MRP study; GS1 2027. GTIN-only resolution now, batch resolution when AI 10 appears.
Shop-management mode inside the one POS application, role-gated, park-first, audited	Single-device shops (September decision); one package for ADR-24 ring updates.
Cross-session refunds, new orders, card tenders, master-data changes are blocked offline, not queued	H-03 as confirmed in the September review (FR-POS-050 wording fix).
Business date set by day-close, not the clock	A 00:30 sale before Z-report belongs to the previous trading day.

Decision	Rationale
Thumbnails shown where a picture prevents a wrong pick: search results, code-lookup confirmation for unbarcoded items, the MRP picker, GRN and count lines	Images are cached read-only from the catalogue snapshot; absent offline means the tag icon shows. Not on receipts or the customer display in phase 1.

12.4 Deliberately not built

- No card processing, no payment gateway, no wallet integration (ADR-19).
- No customer-facing self-checkout, home delivery or online ordering (H-14).
- No till-to-till communication; no shop server; each till syncs to central independently (the primary till is only a counter holder).
- No per-transaction language switch; no cashier discretionary discount of any kind.

12.5 Risks, prototypes and review focus

- Prototypes feeding this design (Stage 3): local database and counter under power loss; Sinhala/Tamil raster printing; 2D scanning; sync at the 15,000-till envelope; two rounds of cashier usability with paper-mirroring screens (T-04).
- Review focus: the offline permission set and the blocked-offline list must be exactly the doc 19 offline_allowed flags; the sync contract's sequence and acknowledgement semantics must be proven under simulated loss, duplicate and reorder of batches.
- Risk: ARM/Android terminal selection is a hardware decision with software consequences (scanner SDK, printer driver, secure storage for PIN hashes and device credential); doc 14 must settle the platform before week 3.

13. M7 – Customers & Payment Recording

Design document: 27 – Customers and Payment Recording Design. **Estimated design effort:** 5 design-days.

The registered customer (a retail-credit concept, not a co-operative member), the digitised khata: per-MPCS credit accounts with limits, balances, repayments and ageing; the tender vocabulary at the till; and payment-receipt documents on the customer side. Cooperative membership, shares, patronage and governance are out of scope by decision.

13.1 Ownership

Owned aggregates and tables (doc 18)	Capabilities (SRS identifiers)
customer, customer_account (per MPCS), customer_phone_history; document type PRC (customer side) and receipt tender lines; ageing projection	FR-CUS-010–060 (global identity, phone lookup with audited change and reuse detection, per-MPCS account, limit and balance, repayments, statements, minimal PII, preferential price hook P3); FR-PAY-010, 020, 060–090 (tender types recorded not processed, customer credit tender, payment receipts, banking record, reconciliation later); I-01, I-02, I-04; K-01 to K-08; ADR-15, ADR-16, ADR-27; PDPA

13.2 Interfaces

Commands accepted	Queries served	Events published
RegisterCustomer, ChangePhone (audited, reuse check), OpenAccount(mpcs, limit), AmendLimit, TenderOnAccount (from receipt), RecordCustomerPayment (PRC), RecordBanking, IssueCustomerStatement, CloseAccount (balance zero)	LookupCustomer(phoneId) scoped to MPCS accounts, GetAccount(balance, limit, ageing), ListOverdue(mpcs), GetStatement	customer.registered/phone_changed.v1, account.opened/limit_amended/closed.v1, account.charged/credited.v1, banking.recorded.v1

Depends on: M1 (MPCS scope), M4 (payment receipt base and SETTLES pattern), M6 (tender lines; offline account tender is allowed only against the cached balance with a configurable offline exposure cap – decision request).

13.3 Key design decisions

Decision	Rationale
One global customer identity, one account per MPCS	ADR-15: a customer who trades with two societies has two independent credit relationships; phone is a lookup attribute, not the identity.
Minimal PII; NIC optional and only when credit is granted	ADR-27, K-08, PDPA minimisation; the audit log never holds values.
Khata tender recorded at the till; limits warn, never block, unless the MPCS configures a hard limit	I-02 is often the largest receivable a society carries; entity policy decides the control (configuration register).
Customer payments settle specific charges, mirroring I-07	Ageing and disputes need it; statements are the customer-facing report.

13.4 Deliberately not built

- No loyalty, points, membership, share capital, patronage, AGM records (K-01, K-04 to K-07).
- No bank or payment integration (R-05); banking is a recorded fact.
- No vouchers or subsidy instruments until later phases (I-01).

13.5 Risks, prototypes and review focus

- Review focus: the PDPA posture (purpose, retention, subject access) documented per field; phone reuse detection flow with real number-recycling cases.
- Decision request: offline account-tender cap and behaviour when the cached balance is stale.

14. M8 – Reporting Pipeline

Design document: 28 – Reporting Pipeline Design. **Estimated design effort:** 6 design-days.

The pipeline, not the reports: projections from domain events into a read replica, end-of-day aggregates, per-entity data-freshness stamps on every federation-level figure, spreadsheet export, and the shared A4 rendering pipeline. Individual reports are specified after the pending report-catalogue workshop and slot into this pipeline.

14.1 Ownership

Owned aggregates and tables (doc 18)	Capabilities (SRS identifiers)
projection tables in a reporting schema (sales by shop/sku/day, stock position, exposure, loss by category, till variance, exception queues), freshness table per location, export jobs, A4 renderer	FR-REP-010–050 (near-real-time operational views, end-of-day aggregates, freshness stamps, spreadsheet export, printed reports); O-01 to O-07; M-04, M-06 (full view-only visibility, no aggregate-only tier); F-09; N-06 (exception queues); P-03

14.2 Interfaces

Commands accepted	Queries served	Events published
RebuildProjection(name, from_seq), RunEndOfDay(business_date), ExportSpreadsheet(report, scope, period), RenderA4(report, scope, language)	GetDashboard(scope) with freshness, GetReport(catalogue id, parameters) with freshness, GetFreshness(entity location)	projection.rebuilt.v1, end_of_day.completed.v1, export.ready.v1

Depends on: All modules' events via the outbox; M1 for scope and RLS on projections; doc 19 event model.

14.3 Key design decisions

Decision	Rationale
Projections are rebuildable from the outbox; the reporting schema is disposable	Doc 19 section 6.5; a wrong projection is rebuilt, never patched.
Every federation-level figure carries the freshness of its least-recent contributor	O-03: some shops are always offline; a figure without a freshness stamp is misleading.
Read replica in phase 1; columnar store when end-of-day exceeds its window	Doc 19 DR-3; provider-neutral.
Decision-driven catalogue, fixed set, spreadsheet export; no report builder	O-01, O-05.
Offline shop reports are the till's own (Z-report, day sales, session), not this pipeline	O-02.

14.4 Deliberately not built

- No self-service report builder; no mobile dashboard application (responsive web suffices, O-06).
- No anonymised aggregate tier (M-06).

14.5 Risks, prototypes and review focus

- Dependency: the report-catalogue workshop (targeted week 3) supplies the fixed set; the pipeline design proceeds without it.
- Review focus: freshness semantics under partial sync; projection idempotency under replay.

15. M9 – Integration

Design document: 29 – Integration Design. **Estimated design effort:** 5 design-days.

The outward surface: an OpenAPI-specified, OAuth2-secured API whose first consumer is the future accounting system; the journal-posting mappings that make that integration a mapping exercise; readiness for Inland Revenue e-invoicing (immutable invoices, disciplined numbering, a reserved integration point); and configurable notifications by SMS and email.

15.1 Ownership

Owned aggregates and tables (doc 18)	Capabilities (SRS identifiers)
api_client, api_scope grants, notification_template, notification_log, posting mappings (data on document_type), e-invoice submission stub	FR-BIL-110 (e-invoice readiness P1, integration P3), FR-BIL-140 (journal mapping per document type); R-05 to R-09 (no bank integration, no portal now, OpenAPI with OAuth2/OIDC, SMS/email notifications, no inherited devices); J-02, J-04; Q-10 (update distribution is M6/ops, referenced only)

15.2 Interfaces

Commands accepted	Queries served	Events published
RegisterApiClient(entity scope), IssueClientCredential, PublishNotificationTemplate, SendNotification (from events), ExportJournal(entity, period) using posting maps, SubmitEInvoice (stub, P3)	OpenAPI read endpoints over documents, stock, exposure and statements, scoped by client entity and the same RLS classes; GetPostingMap(doc_type)	notification.sent/failed.v1, journal.exported.v1, api_client.registered.v1

Depends on: All modules' queries and events; M1 for client scope; doc 19 identity for OAuth2.

15.3 Key design decisions

Decision	Rationale
OpenAPI is the contract; REST only	R-07; one expected consumer; no GraphQL (Section 4).
Journal mappings live as data on document types	J-02: the accounting choice is deferred; mapping is a configuration exercise when it lands.
E-invoice readiness is a reserved seam, not a live integration	J-04: immutability and numbering are already in doc 18; the submission adapter is phase 3.
Notifications are event-driven templates in the shop language with English fallback	R-08; dispatch alerts and statements first.

15.4 Deliberately not built

- No bank files, no government portal, no marketplace or e-commerce connectors (R-05, R-06, H-14).
- No inbound write API in phase 1 beyond the accounting consumer's read needs.

15.5 Risks, prototypes and review focus

- Review focus: API scope enforcement uses the same RLS classes; rate limits per client; the posting map must be reviewed by an accountant against the trading-document state machines.
- Decision request: SMS gateway selection is a procurement item, not a design item; the adapter interface is designed provider-neutral.

15A. M10 – Procurement (phase 2 boundary)

Design document: Phase 2: design document to be numbered when scheduled. **Estimated design effort:** not estimated (phase 2).

Not designed in phase 1; placed on the map so that phase-1 modules leave the seams it needs. The blueprint here is a boundary statement, not a design: what M10 will own, which phase-1 fields exist for it, and what it must not pull into phase 1.

15A.1 Ownership

Owned aggregates and tables (doc 18)	Capabilities (SRS identifiers)
Supplier master (takes over M2's minimal supplier table); supplier contracts and agreed prices; quotation register with audited award; purchase order; market price observations by source; shipment and landed cost; supplier performance; demand-forecast inputs	FR-PRO-010–060 (all P2 except the P3 Federation repacking permission flip); L-01 to L-07; ADR-06

15A.2 Interfaces

Commands accepted	Queries served	Events published
RegisterSupplier, RecordQuotation, AwardQuotation, IssuePurchaseOrder, RecordShipment, FinaliseLandedCost, RecordMarketPrice (all P2)	ListSuppliers, GetPurchaseOrder, GetExpectedReceipts(warehouse), GetSupplierPerformance (all P2)	purchase_order.issued.v1, quotation.awarded.v1, landed_cost.finalised.v1, supplier_performance.computed.v1 (P2)

Depends on: M1, M2, M4 (Federation warehouse GRNs cite a purchase order), M5 (lot cost restatement from landed cost), M8 (sales aggregates for forecasting).

15A.3 Key design decisions

Decision	Rationale
Phase-1 seams left for it	M2 keeps a minimal supplier table that M10 later absorbs; M4's Federation warehouse GRN carries an optional purchase_order_ref; M5's lot cost is restatable by an event; M8's aggregates are queryable by SKU, region and season.
Price forecasting is a scenario tool	L-02: never an automated decision-maker; the module produces comparisons, people decide.
No public tender portal	L-04: a quotation register with an audit trail, not a marketplace.

15A.4 Deliberately not built

- Nothing in phase 1 beyond the seams above; no supplier screens, no purchase orders, no landed cost.

15A.5 Risks, prototypes and review focus

- Risk: if the seams are not left now, phase 2 starts with a data migration inside M2 and M4. The seams are cheap; check them at the M2 and M4 contract freezes.

16. Review gates, decision requests and exit criteria

16.1 Gates

Gate	When	Passes when
Contract freeze (per module)	End of the module's review week	Commands, queries and events published with versions; at least one dependant has acknowledged
State-machine reconciliation	M4 and M5 reviews	Every doc 03 transition mapped; every transition has permission, reason rule, offline flag, audit event
Prototype incorporation	Weeks 2–6 as spikes complete	Each spike's findings recorded as an ADR or a design change; none left in a spike report
Cross-module reconciliation	Week 6	No two modules claim one table; all doc 18/19 change requests raised; no unresolved contract conflicts
Stage exit	End of week 6 (or 7)	All nine designs at definition of done; decision requests assigned; Stage 4 inputs (hosting, platform) available

16.2 Decision requests raised by this plan

#	Point	Proposed	Decider	Needed by
DR-1	Trading documents as a ninth module (M4) rather than split across inventory and billing	Closed: adopted in doc 17 v0.2.1 (ADR-37); Procurement kept as M10 (P2)	–	–
DR-2	Invoice-on-dispatch as a relationship term in phase 1	Defer to phase 2; invoice follows GRN in phase 1	Federation admins	Week 4 (M4 review)
DR-3	Offline khata tender cap and stale-balance behaviour	Allow up to a configurable cap against cached balance; warn	Federation admins with MPCS treasurer input	Week 4 (M7)
DR-4	POS platform (ARM/Android terminal model, scanner SDK, printer)	Settle in doc 14	Engineering lead	Week 3 (M6 start)
DR-5	Report catalogue workshop date	Week 3	Federation	Week 3
DR-6	Tax advisor answers (numbering FY reset, GRN fiscal scope, multi-series)	Assumptions per doc 18 until answered	Tax advisor	Week 4
DR-7	Design team allocation (2 vs 3 module designers)	3, to keep Track B independent	Chamin	Week 1

16.3 What Stage 2 hands to Stage 4 and 5

- Nine module design-and-guide pairs with published contracts, committed DDL, seed data and OpenAPI slices, and a consolidated change-request list against docs 18 and 19.
- A consolidated configuration register (from each module's section 7) that becomes the onboarding playbook's input.
- A consolidated test strategy with the doc 13 scenarios allocated to modules and the property-based invariants named.
- The till platform decision, prototype results and the sync contract, which Stage 4 turns into deployment, update and support design.
- An updated decision log: every DR closed or carried with an owner.

Appendix A. Register of deferred capability

What the simplifications and phase boundaries leave out of version 1, stated so that nothing is lost by default. Reviewed at every phase gate. Class N: no functional loss (mechanism choice). Class D: deferred with a designed path back. Class C: consequence accepted knowingly. Class B: out by the requirements baseline, listed with the cost of reversal.

Item	Class	What v1 does instead	Path back and cost
Independent scaling per module	N	One deployable; a second instance with an ingest-only role for sync bursts	Extract a module; boundaries are enforced so extraction is mechanical. Moderate.
Per-entity custom approval chains	D	Fixed state machines with value-threshold role parameters (C-02, E-07, F-04)	Approval-policy table per document type; no engine. Low.
New promotion kinds (mix-and-match, coupons, vouchers, subsidy instruments, segment targeting)	D	Closed rule vocabulary: time-limited, quantity break, bill threshold, free item, expiry markdown	Add rule kinds under the versioned schema; each is a release. Low per kind. Vouchers and subsidy are later phases by answer (I-01).
Ad hoc analysis	D	Fixed catalogue, spreadsheet export; adopted for v1 : read-only SQL access to the reporting replica for a named Federation analyst role under RLS	Report builder remains backlog (O-05).
As-of queries on master data (SKU attributes, roles, relationships)	D	Effective-dated prices, tax and control prices; immutable documents; audit before/after diffs for the rest	Versioned rows on sku and role if needed. Moderate.
Warehouse operations (bins, put-away, pick paths, waves)	D	Allocation produces a pick list; one stock location per warehouse	WMS module on the same movement ledger, phase 2/3. Moderate to high; the most likely operational pressure point at 75 distributors.
Custom fields on SKU and customer	D	Adopted for v1 : bounded, entity-owned attributes jsonb (non-searchable, no behaviour) on sku and customer	Searchable or behavioural attributes need proper columns. Low.
Product images	–	Adopted for v1 (M2 ownership, M6 display): thumbnails on till lookups, picker, GRN and count lines	Full-size images, images on customer display: phase 2. Low.
Tills in one shop are independent while offline	C	Each till deducts its own cached lot; negative stock permitted and flagged on sync (E-09); parked sales are per till	Shop-hub role on the primary till (LAN peer sync). Moderate. Revisit if usability rounds or pilot show it bites.
Offline changes limited to what a till issues	C	One writer per record type (ADR-10, H-03): no offline catalogue edits, orders, user changes	None intended; this is the price of having no merge logic.
Broker-level replay and streaming consumers	N	Outbox retains; RabbitMQ/NATS buffers and fans out	Kafka if broker-level replay is ever required. Moderate.
Per-tenant isolation (performance, export-by-drop)	N	Shared rows with RLS; societies cannot leave (M-07)	Not needed.
Shelf versus back-store	B (E-02)	One stock location per shop	Sub-location on stock_lot and replenishment tasks. Moderate.
Serialised items	B (B-05)	None	Per-unit tracking table and scan flow. High.
Returnable containers	B (D-10)	None	Container balance per shop as a non-stock document. Moderate.
Cold-chain telemetry	B (E-10)	Physical handling only	Sensor integration. Low to moderate.
Membership, loyalty, patronage, shares, governance	B (K-01 to K-07)	Registered customers only	Large; a separate programme.
Home delivery, online ordering, self-checkout	B (H-14)	None	Channel module. High.
Scheduled future price lists	B (G-08)	Apply on propagation with apply_from	Effective-date columns already exist. Low.
Invoice on dispatch	B (doc 20 DR-2)	Invoice follows GRN	Relationship term plus DN-referenced invoice. Low.

17. Traceability summary

Module	Requirement families
M1	FR-PTY, FR-SEC, FR-GOV; A, M, N sections; ADR-18, 26
M2	FR-SKU, FR-LOC-020; B section; ADR-07, 08; T-03
M3	FR-PRC; G section; F-07; ADR-14; multi-MRP and control-price decisions
M4	FR-ORD, FR-DEL, FR-BIL; A, C, D, I-05 to I-07, J sections; ADR-01, 03, 11, 12
M5	FR-INV, FR-LOSS; E, F sections; B-06; R-02; ADR-07
M6	FR-POS, FR-PAY (till subset); H, I-03, O-02, P-05, Q-07 sections; ADR-10, 19, 24
M7	FR-CUS, FR-PAY (customer subset); I-01, I-02, I-04, K section; ADR-15, 16, 27
M8	FR-REP; O section; M-04, M-06; N-06
M9	FR-BIL-110, 140; R section; J-02, J-04
M10 (P2)	FR-PRO; L section; ADR-06; doc 17 §7.10

End of document. Draft for design review; module blueprints are superseded by their detailed designs (docs 21–29) as each reaches definition of done.